

# Chilled Water Cooling Coil Design Report - v2.3

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Physical flow geometry: CROSS-FLOW (air perpendicular to tube/coolant direction). Water enters air-leaving side (cross-counterflow tendency)

## Design Inputs

| Input / construction                | Value                                                      |
|-------------------------------------|------------------------------------------------------------|
| Face W x H                          | 1.200 x 0.850 m                                            |
| Number of tube rows (airflow depth) | 6                                                          |
| Nominal tube-bank depth             | 132.0 mm                                                   |
| Fin type                            | Plain fin                                                  |
| Fin material                        | Aluminum                                                   |
| Tube material                       | Copper                                                     |
| FPI / fin pitch                     | 10.0 1/in / 2.540 mm                                       |
| Fin thickness                       | 0.120 mm                                                   |
| Tube OD / wall                      | 9.530 / 0.350 mm                                           |
| Pt / PI                             | 25.400 / 22.000 mm                                         |
| Airflow                             | 2.0400 m3/s = 7344 m3/h = 4323 CFM                         |
| Gross face velocity                 | 2.000 m/s                                                  |
| Entering air DB / RH                | 27.00 degC / 50.0%                                         |
| Coolant / concentration             | Water / 0.0%                                               |
| Coolant inlet / pressure            | 7.00 degC / 300.0 kPa abs                                  |
| Coolant mass flow / circuits        | 1.2000 kg/s / 6                                            |
| Circuit tube-end arrangement        | Same tube end (even passes/circuit required)               |
| Calculated tubes / row / total      | 8 / 48                                                     |
| Circuit model                       | Explicit user-defined routed circuits                      |
| Water row progression               | Water enters air-leaving side (cross-counterflow tendency) |

## Performance

| Performance          | Value                    |
|----------------------|--------------------------|
| Total capacity       | 48.00 kW                 |
| Sensible / latent    | 34.00 / 14.00 kW         |
| SHR                  | 0.708                    |
| Leaving air DB / WB  | 13.00 / 12.00 degC       |
| Leaving air RH       | 90.0%                    |
| Leaving coolant      | 13.60 degC               |
| Condensate           | 20.00 kg/h               |
| Wet fraction / state | 90.0% / Fully wet by row |
| Air dP               | 120.0 Pa                 |
| Water dP avg         | 35.50 kPa                |
| Water dP min / max   | 35.00 / 36.00 kPa        |

## Heat Transfer Diagnostics

| Heat-transfer diagnostic            | Value                |
|-------------------------------------|----------------------|
| C air                               | 2.400 kW/K           |
| C coolant                           | 5.000 kW/K           |
| Cmin / Cmax                         | 2.400 / 5.000 kW/K   |
| Capacity ratio Cr                   | 0.480                |
| Dry-reference NTU                   | 2.100                |
| Dry cross-flow effectiveness        | 0.750                |
| Wet enthalpy effectiveness          | 0.680                |
| Air temperature effectiveness       | 0.740                |
| Air Re / Pr                         | 2000 / 0.710         |
| Coolant Re / Pr                     | 12000 / 8.000        |
| Capacity-rate limiting side         | Air side             |
| Resistance limiting side            | Air side             |
| Resistance split air / water / wall | 75.0% / 24.0% / 1.0% |

## Velocities

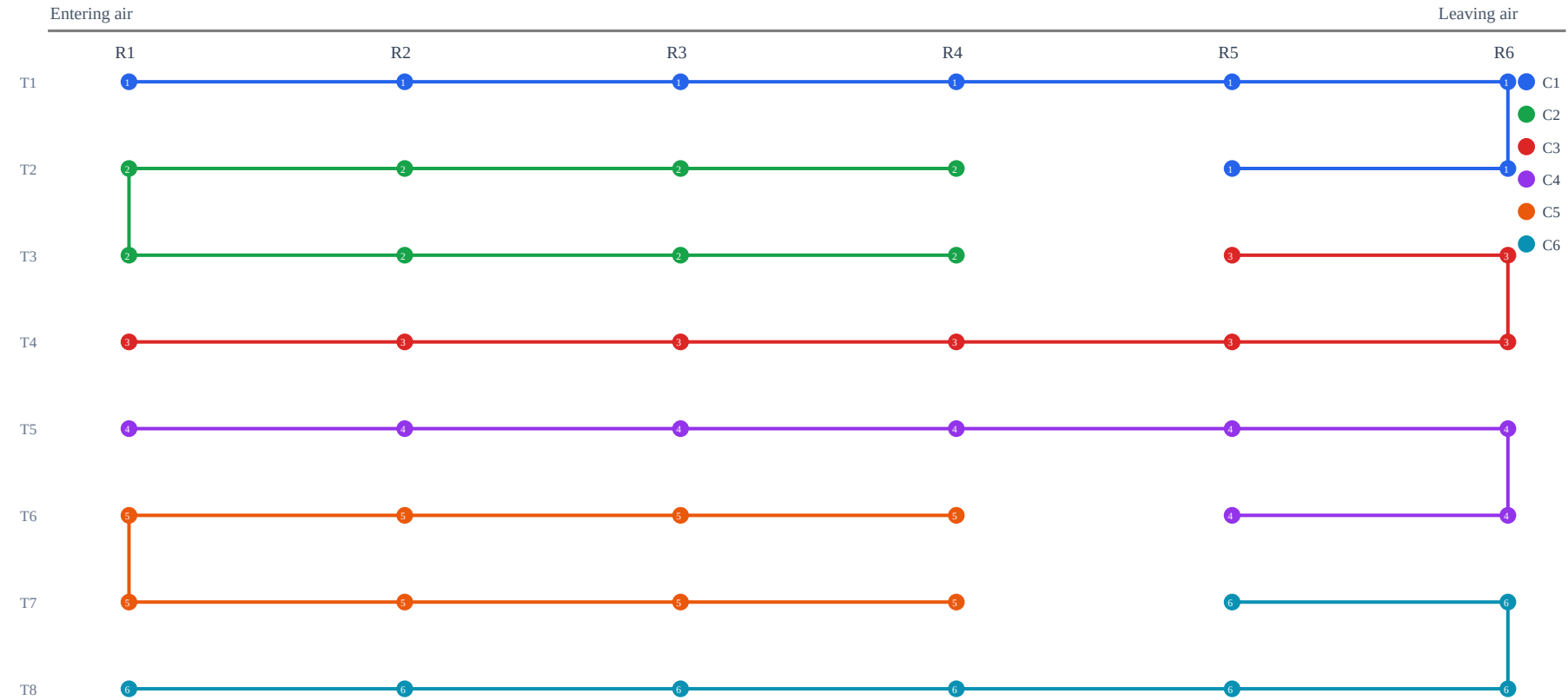
| Velocity / flow item                | Value       |
|-------------------------------------|-------------|
| Gross air face velocity             | 2.000 m/s   |
| Max air velocity between fins/tubes | 3.200 m/s   |
| Coolant velocity in one tube        | 1.200 m/s   |
| Supply header velocity              | 1.100 m/s   |
| Return header velocity              | 1.100 m/s   |
| Coolant mass flow per circuit       | 0.2000 kg/s |

## Calculated Geometry

| Calculated geometry            | Value                                        |
|--------------------------------|----------------------------------------------|
| Tube rows in airflow direction | 6                                            |
| Nominal tube-bank depth        | 132.0 mm                                     |
| Tubes / row                    | 8                                            |
| Total tubes                    | 48                                           |
| Selected circuits              | 6                                            |
| Circuit tube-end arrangement   | Same tube end (even passes/circuit required) |
| Circuit model                  | Explicit user-defined routed circuits        |
| Tube length                    | 1.200 m                                      |
| Fin count                      | 472                                          |
| Face / free-flow area          | 1.020 / 0.610 m <sup>2</sup>                 |
| Free-area ratio                | 0.600                                        |
| Air-side area                  | 140.00 m <sup>2</sup>                        |
| Inside tube area               | 1.30 m <sup>2</sup>                          |
| Air-side correlation           | Wang-Chi-Chang 2000 plain fin                |

# Physical Water Circuiting

Tube IDs use R# for row in the airflow direction and T# for vertical tube position. Lines connect the ordered straight tube passes; each connection represents a return bend at a tube end.



## Circuit hydraulic distribution

| Circuit | Passes | Inlet | Outlet | Flow kg/s | Tube v m/s | Flow dev % | Total dP kPa |
|---------|--------|-------|--------|-----------|------------|------------|--------------|
| 1       | 8      | R1-T1 | R5-T2  | 0.2000    | 1.030      | +0.0       | 35.20        |
| 2       | 8      | R4-T2 | R4-T3  | 0.2000    | 1.060      | +0.0       | 35.40        |
| 3       | 8      | R5-T3 | R1-T4  | 0.2000    | 1.090      | +0.0       | 35.60        |
| 4       | 8      | R1-T5 | R5-T6  | 0.2000    | 1.120      | +0.0       | 35.80        |
| 5       | 8      | R4-T6 | R4-T7  | 0.2000    | 1.150      | +0.0       | 36.00        |
| 6       | 8      | R5-T7 | R1-T8  | 0.2000    | 1.180      | +0.0       | 36.20        |

## Circuit outlet temperatures

| Circuit | Passes | Flow kg/s | Water out degC | Circuit Q kW |
|---------|--------|-----------|----------------|--------------|
| 1       | 8      | 0.2000    | 13.45          | 8.00         |
| 2       | 8      | 0.2000    | 13.50          | 8.00         |
| 3       | 8      | 0.2000    | 13.55          | 8.00         |
| 4       | 8      | 0.2000    | 13.60          | 8.00         |
| 5       | 8      | 0.2000    | 13.65          | 8.00         |
| 6       | 8      | 0.2000    | 13.70          | 8.00         |

Row-duty-conserving circuit temperature post-processor; not yet a fully coupled 2-D circuit thermal solve.

Row-by-Row Thermal Marching

Rows are numbered in the air-flow direction. Physical geometry is CROSS-FLOW at every row. Water enters air-leaving side (cross-counterflow tendency). Converged: True in 12 iteration(s).

| Row | Air DB in | Air DB out | Air WB out | RH out % | Water in | Water out | Q kW | Latent kW | Wet % | Mode      |
|-----|-----------|------------|------------|----------|----------|-----------|------|-----------|-------|-----------|
| 1   | 27.00     | 24.70      | 19.00      | 65.0     | 12.00    | 13.00     | 8.00 | 2.00      | 80.0  | Fully wet |
| 2   | 24.70     | 22.50      | 17.80      | 70.0     | 11.20    | 12.20     | 8.00 | 2.00      | 80.0  | Fully wet |
| 3   | 22.40     | 20.30      | 16.60      | 75.0     | 10.40    | 11.40     | 8.00 | 2.00      | 80.0  | Fully wet |
| 4   | 20.10     | 18.10      | 15.40      | 80.0     | 9.60     | 10.60     | 8.00 | 2.00      | 80.0  | Fully wet |
| 5   | 17.80     | 15.90      | 14.20      | 85.0     | 8.80     | 9.80      | 8.00 | 2.00      | 80.0  | Fully wet |
| 6   | 15.50     | 13.70      | 13.00      | 90.0     | 8.00     | 9.00      | 8.00 | 2.00      | 80.0  | Fully wet |

## Method / Validation Note

- The physical fin-and-tube coil is cross-flow: air moves through the depth and coolant moves along the tubes, approximately 90 degrees to the air velocity.
- The water-row-progression option only changes which coil-depth side receives cold water. It is not a physical parallel-flow/counterflow change of the local air/tube orientation.
- Plain fins use the Wang, Chi & Chang (2000) plain fin  $j/f$  correlation. Wavy+louvered fins use the Wang-Tsai-Lu correlation as documented by ACHP.
- Wavy-only mode is deliberately labelled a calibration-required baseline rather than inventing unverified correlation coefficients.
- Water-side heat transfer uses Gnielinski with transition treatment; tube/header pressure loss uses Darcy-Weisbach with Churchill friction.
- Wet-coil capacity uses a segmented enthalpy-potential dry/part-wet/wet treatment. v2.3 can use an explicit physical circuit routing map for circuit-resolved hydraulics and a row-duty-conserving tube-by-tube coolant-temperature post-processor.
- The v2.3 tube-temperature post-processor does not yet feed individual circuit temperatures back into each local tube heat-transfer calculation; that fully coupled 2-D circuit thermal solve is a further refinement.
- Validate air HTC, wet air dP and fitting K values against actual coil test or trusted manufacturer data before production release.